FACULTEIT INGENIEURSWETENSCHAPPEN
EN ARCHITECTUUR

INTERMEDIATE REPORT – ASSIGNMENT 5

COMPUTATIONAL BIOFLUID MECHANICS

ACADEMIC YEAR 2025-2026

Name student

Prof. Charlotte Debbaut, Prof. Patrick Segers

Dr. ir. Mohammad Rahimi Gorji, ir. Saar Vermijs, ir. Jessie Duquesne,
ir. Kaat Wille, ir. Irea Vervaeke

UNIVERSITEIT
GENT

ASSIGNMENT 5: MESH SENSITIVITY STUDY OF POST-OP CASE

Submit individually the .cas and .dat file of the steady simulation with your final mesh (Fluent) to Assignment 5 on Ufora. In addition, discuss/show the following points below. Make sure that figures/tables are high-quality and complete, that they have numbers and clear captions, and that they are referred to in the text.

Results (/4) *(max. 150 words)*

- *Please plot the results of your mesh sensitivity analysis using an X-Y plot. On the X axis, report the number of mesh elements. On the Y-axis, plot your parameter(s) of interest. Repeat for every parameter of interest. (/2)*
- *Calculate and plot the percentual difference between values of your parameter of interest for the different meshes by comparison with the densest mesh (i.e. compare the percentual difference in the parameter of interest between each mesh and your densest mesh). Use a formula to indicate how exactly you calculated the percentual difference. (/2)*

Discussion (/2)

Explain how you would interpret the results above. What is your optimal mesh? (/2) (max. 100 words).

Limitations (/2)

- *Discuss limitations of the current modelling approach. How can you make the current model more realistic? (/1) (max. 150 words)*
- *Given more time and computational resources, how would you expand upon this mesh sensitivity study? (/1) (max. 100 words).*

Interpretation (/2)

Assume that we apply a time-varying velocity profile at the inlet of the post-op geometry, corresponding to an increasing velocity during inspiration. Let's also focus on the part of the post-op geometry before the bifurcation, and simplify this to a tube of constant diameter. Knowing the definition of resistance (Assignment 3) and the relationship between pressure drop and flow rate in this tube of constant diameter (Assignment 3), can you explain what the time-dependent resistance would look like through this period of inspiration? Please draw/explain (max. 100 words) (/2)